LedLamp — rebuilding a Chinese ceiling lamp as a Zigbee tunable-white light

Keep the lamp's **metal case** and diffuser; replace everything inside with:

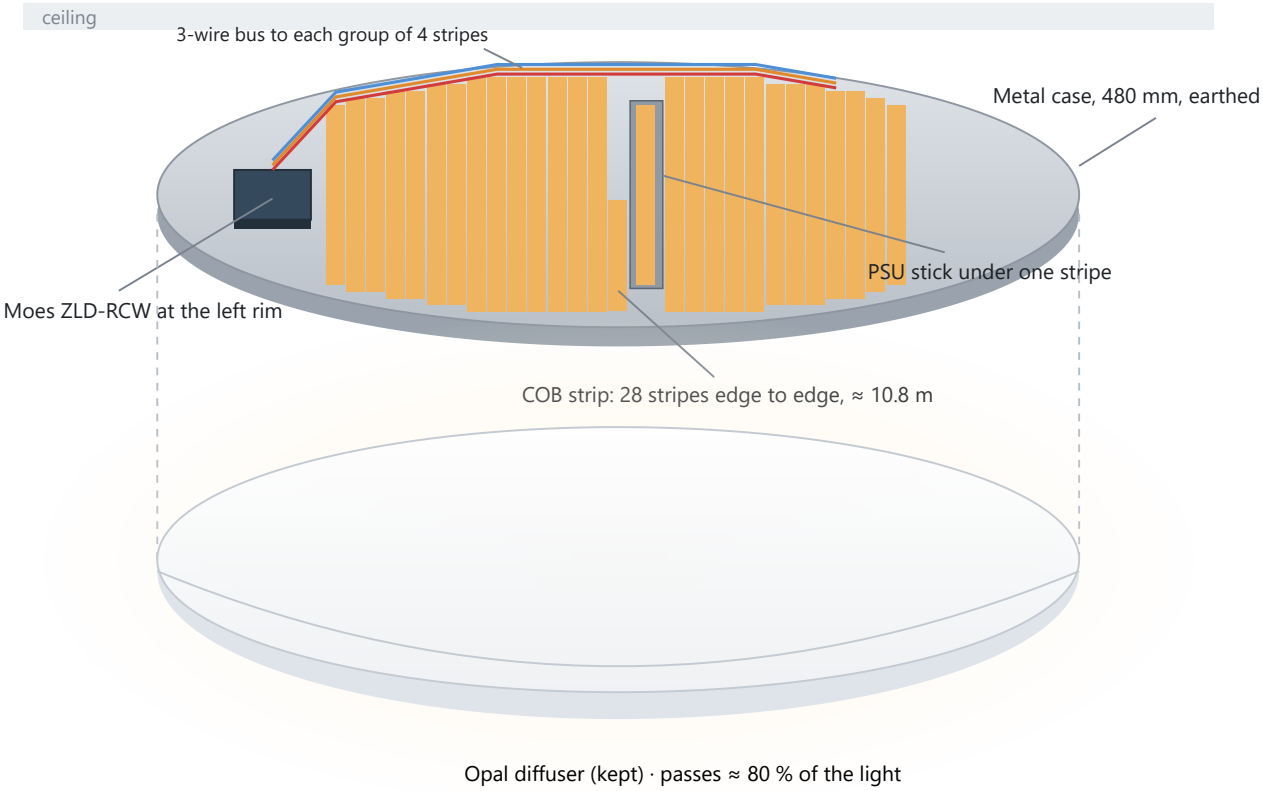
1. **COB LED strip, tunable white (CCT)**: warm 2700 K to cool 6500 K, dot-free under the diffuser.
2. **Zigbee CCT controller**: paired with the house's Zigbee2MQTT, so HA gets brightness + colour temperature.
3. **Slim 24 V power supply**.

Constraints: living room (IP20 parts are fine, nothing sealed needed), **Zigbee only**, **≤ 2500 UAH** in total, and as bright as possible within that.

The lamp: Luxel **CLRR-72**, a round remote-controlled fitting, **480 mm across** (measured) and about 50 mm deep. Its label says **72 W, 175–260 V, 5400 lm**.

What's inside now (from a photo of the opened lamp): two square LED boards, about 33 cm and 24 cm across, carrying roughly **75 separate LEDs under small lenses**, about 2.3 m of board in total. A small mains driver in a clear plastic box and a 3-way terminal block (L, N, earth) sit in the middle, and the base is painted white metal. There are mounting holes near the rim at the top, bottom and right. The separate lensed LEDs are why the old light looks spotty.

Prices and stock were checked on **2026-09-26** and re-checked on **2026-10-03**, when every part was ready to ship on prom.ua; the power supply was re-picked on **2026-10-11** for the dense layout. They change often. Rozetka refused automated lookups (HTTP 403). A printable version with drawings, `LedLamp.pdf`, is built from this README and `docs/` by `tools/build_pdf.py`.



1. Parts to buy: ≈ 4200 UAH for the lamp, ≈ 820 for the wall switch

#	Part	Pick	Shop	Price, UAH
1	CCT COB strip, **13 m** (≈ 12 m used, 1 m spare)	**LEDTech 24V COB/FCOB CCT 2700–6500K Multi White**: 14 W/m (7+7), 1400 lm/m, 608 LED/m, Ra>90, **10 mm** wide	prom.ua (LEDTechnics)	~2930 (225.40/m × 13 m)
2	Zigbee controller	**Moes ZLD-RCW** RGB+CCT, with R, G and B left empty: Zigbee 3.0 (Tuya), DC 12–24 V, 5 channels, 15 A, 2700–6500 K, **90×40×24 mm**	prom.ua (SELLBOT), in one order with the wall switch	849
3	Power supply	**Diasha PPF-SL-C-24V-120W** "pencil" stick: 24 V / 5 A, 120 W, **340×17×18 mm**, metal case, IP20, passive cooling	prom.ua (EcoLed), Smart delivery (free to a Nova Poshta branch)	320
4	Wire (0.75 + 0.5 mm²), heat-shrink, 3 **WAGO 221-412** (2-wire, genuine, 23 each)	wire local; WAGO from prom.ua (Електро Крамниця), in one order with the wall switch's blank	—	~100
	Total (lamp)			**≈ 4200**

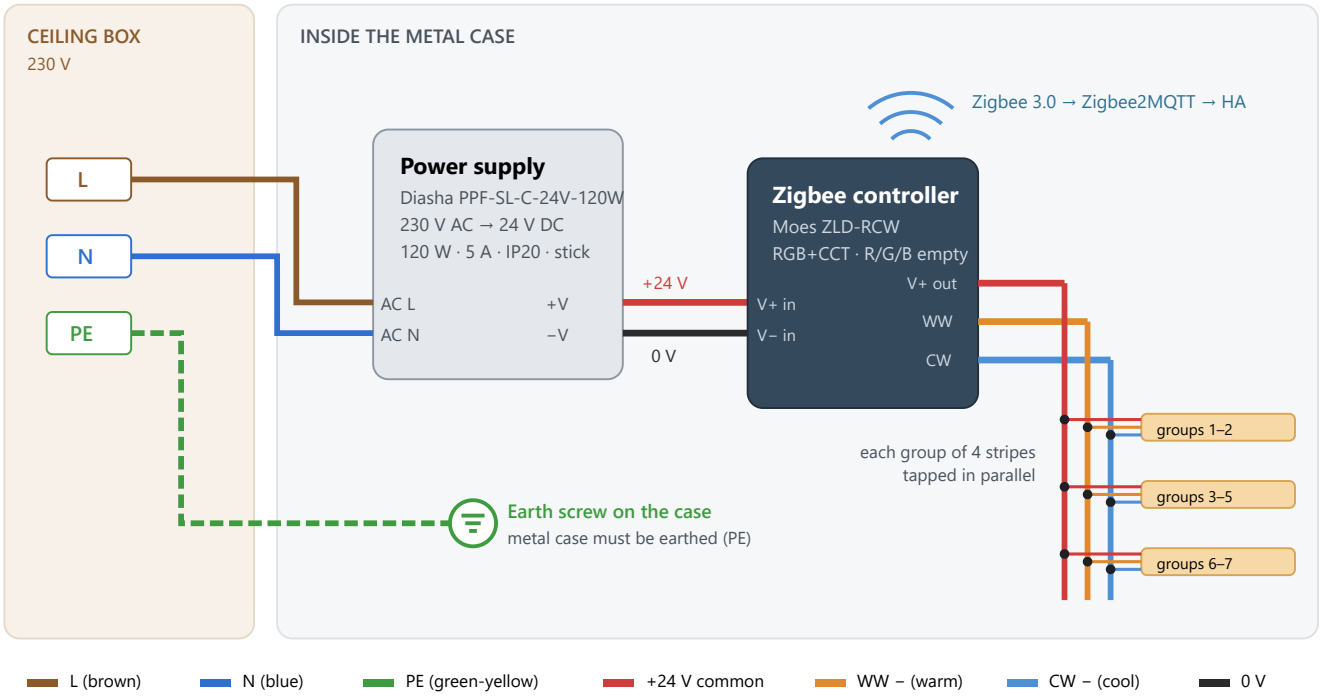
About 13 m of strip. The dense layout (28 stripes, ≈ 10.8 m; up to 30 if your case allows, ≈ 11.9 m) leaves the rest for cutting mistakes and a spare stripe for repairs. If the first 7 m are already bought, about 6 m more (~1350) is enough. The listing is priced per metre; if they only sell whole 5 m reels, three reels (15 m) cost ~3380. The extra strip and the bigger power supply put the lamp well over the original 2500 budget, in exchange for about twice the light; the wall switch (`docs/wall-switch.md`) adds ≈ 820. The LT COB-24-MW-608 at svetum (same 7+7 W/m, CRI 90, 10 mm) is sold only in 5 m steps.

Why these: - **Strip:** the only one found that meets every point (2700–6500 K, Ra>90, 10 mm, 14 W/m) at the lowest price. Light per metre decides the brightness, so stay at 14 W/m: 10 W/m strips give about a third less. - **Controller:** no CCT-only Zigbee controller cheaper than the E2-ZR (1210) was in stock in Ukraine on 2026-10-03 (the Tuya C02Z at 600 was sold out), so the pick is an RGB+CCT unit with R, G and B left unused. The Moes comes from SELLBOT, a shop with returns and a 99 % rating, and travels in the same parcel as the wall switch, so it adds no delivery. Zigbee2MQTT supports it (ZLD-RCW_1: brightness, color_temp 153–500 mired, colour, `do_not_disturb`, power-on behaviour), and it is mains-powered, so it is a Zigbee router. Its PWM frequency is unpublished (flicker on phone cameras is possible; flicker you can see is not expected) and it has no RF remote. Its temperature rating isn't published either, so it sits at the rim, away from the strip, and the one-hour test (checklist step 9) checks it. - **Power supply:** it has to be a **metal stick**, because one stripe is stuck on top of it. The 28 stripes draw ≈ 76 W (30 stripes ≈ 83 W, 3.5 A), since the controller caps the strip at 7 W/m. The 120 W Diasha runs at about 65–70 % load, where a 100 W stick would be past 80 % in a warm case with a stripe heating it, and BIOM's sticks stop at 72 W. It comes with prom.ua Smart delivery. The listing names no protections and its spec table says 150 W / 6.25 A against 120 W / 5 A in the title, so **ask the seller in the prom chat for short-circuit and overload protection before paying**. At 340 mm it fits only within about ±13 cm of the centre, where the case is tall enough. All the candidates are in `docs/parts-candidates.md`.

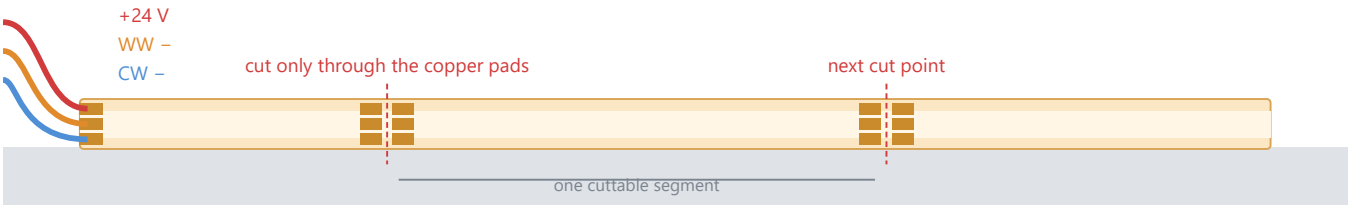
Swaps if something is out of stock: - Power supply: PROLUM Series-UTF 100W 24V at Epicentr (512, free delivery to Epicentr parcel lockers), 264×21.5×20 mm, sealed aluminium, IP67, short-circuit, overload, overheat and overvoltage protection, 24 months: the best-protected stick-shaped unit, at 77–83 % load. Or the LED STORY Profi 100W stick (390, led-story.ua only, no Smart), 317×17×17 mm, metal, overload/short-circuit/overheat protection, 2 years. - Controller: the **MiBoxer FUT037Z+** (OLX, 679, about 807 delivered, Z2M), another RGB+CCT unit wired the same way (set it to RGB+CCT mode with its button). Same maker as the E2-ZR, 16 kHz PWM and an RF remote, 74.5 × 36 × 17 mm, rated −10 to 40 °C; but a private seller, so confirm **FUT037Z+** on the label (the plain FUT037 is RF-only) and pay through OLX Delivery. - Controller, CCT-only: the **MiBoxer E2-ZR** (OPTSVET, 1210, Z2M), the original pick (lamp ≈ 3215). Purpose-built for tunable white, so there is no colour mode to avoid, but at 100.6 × 40 mm it is a tight fit at the rim.

2. How it fits together

Wiring



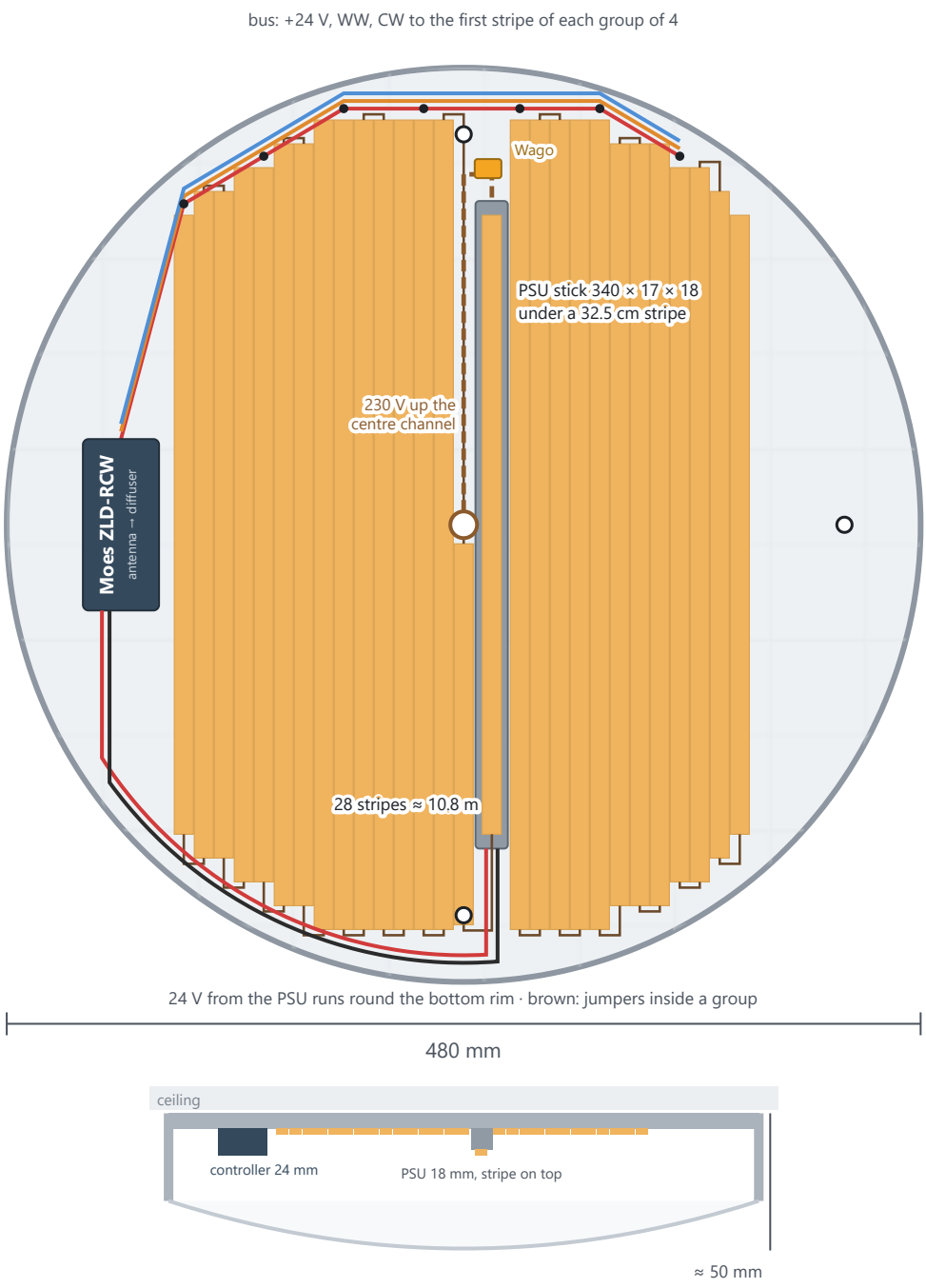
- **The strip is common-anode**: one `+24 V` pad and two negative pads (`WW`, `CW`). The controller switches the negatives. Follow the controller's label for which outputs are V+ and the two whites, and leave R, G and B unconnected (the Moes is RGB+CCT only, so there is no mode to set).
- **Use colour temperature only.** HA also shows a colour picker for this light. Picking a colour drives only R, G and B, which aren't connected, so the lamp goes dark. Nothing is harmed, but the dashboard should offer colour temperature only.
- **Earth the case.** The PSU sits on mains inside a metal case; connect PE to the case's earth screw. The 24 V side is then touch-safe.
- **Wire gauge**: mains 0.75 mm²; PSU → controller and the 3-wire bus 0.75–1.0 mm² (up to 3.5 A in total); tails and jumpers between stripes 0.5 mm² / 20–22 AWG (under 0.5 A per group).
- **Solder the strip.** Clip-on connectors are unreliable on COB strip; tin the pads and solder the three wires.



bare, cleaned metal of the case (isopropyl alcohol)

The pad order (+, WW, CW) varies between strips: read the print next to the pads, not this drawing.
Tin the pads, solder 0.5 mm² tails, cover with heat-shrink or a drop of neutral silicone.

Layout in the 480 mm case: 28 stripes, edge to edge



Strip faces down onto the diffuser. COB needs only
15–20 mm of distance to look dot-free. Height not to scale.

- **28 vertical stripes laid edge to edge** (10.5 mm centre to centre) in a band about 300 mm wide through the middle, each ending about 15 mm from the wall and cut at the nearest pads (every 25 mm): 42.5 cm in the middle down to 32.5 cm at the band's edges, **≈ 10.8 m in total**. If your case has room, one more stripe at each edge makes 30 (≈ 11.9 m, ≈ 83 W); the 120 W supply covers that too.
- **The centre column** carries the mains cable in its top half (from the centre hole up to the PSU's input end) and a 20 cm stripe in its bottom half.
- **The PSU stick sits right of the centre hole**, in place of one stripe, with a **32.5 cm stripe stuck on top** of it. It is 340 mm long, so it fits only within about ±13 cm of the centre. Its input end is at the top, where the mains cable meets it in a Wago 221; its output end is at the bottom, and the 24 V pair runs round the bottom rim to the controller. The stick's metal case is not earthed (2-wire input), so insulate the stripe's solder joints and tails from it like everywhere else.
- **The controller sits in the left rim segment**, outside the band. The Moes is **24 mm** thick, so check the free height under the diffuser (checklist step 2).

- **Check the mounting holes** on your case before sticking anything. In the drawing the top hole falls in the centre channel, the bottom one just below the centre stripe, and the right one in the rim segment.
- **Feed the stripes in groups of 4**: neighbours linked at alternating ends with short jumpers (+, WW, CW), seven groups in all, each fed at the top end of its first stripe from the 3-wire bus. A group is about 1.6 m and under 0.5 A, so the brightness stays even. Do not daisy-chain all the stripes end to end.
- **The old terminal block has to go**; the Wago at the PSU's input end replaces it.
- **Heat:** a Zigbee CCT controller keeps warm + cool at 100 % in total, so each stripe draws at most 7 W/m: ≈ 76 W over the base, ~620 W/m^2 under the band. A test on 2026-10-11 (part of the stripes, edge to edge, one hour at full power) left the strip barely warmer than a hand, so it goes straight onto the cleaned, painted steel. The stripe on the PSU is the one to watch: the supply's own ~10 W of loss heats it from below, so measure it separately in the one-hour test (checklist step 9) and leave it off if it passes ~60 °C.
- **Zigbee through metal is weak.** Mount the controller with the antenna end toward the diffuser. It is mains-powered, so it also works as a **Zigbee router**. Its PWM frequency isn't published, so phone cameras may show banding; the eye shouldn't.
- COB strip needs little distance to the diffuser: 15–20 mm already looks dot-free.

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3. Build checklist

1. **Before the rebuild**, take lux-meter readings under the old lamp at warm, neutral and cool (see `docs/brightness.md`).
2. Measure the free height inside and how the diffuser attaches.
3. Remove the two old LED boards, the driver and the centre terminal block. Clean the painted base where the strip will go (isopropyl alcohol).
4. Mark the centre line, the edges of the ~300 mm strip band, the mounting holes, the centre channel, the PSU column right of the centre hole and the controller spot in the left rim segment.
5. Cut each stripe at the nearest pads so it ends ~15 mm from the wall, stick them on edge to edge, and insulate every solder joint (heat-shrink or Kapton tape) so no pad touches the metal.
6. Mount the PSU stick (thermal tape or screws) and stick its stripe on top; mount the controller at the left rim. Link the stripes in groups of 4, run the 3-wire bus to each group, earth the case, run the mains cable up the centre channel and join it at the PSU's input end with Wago 221.
7. Bench test first on 24 V: both channels light, correct warm/cool order (swap WW/CW if reversed). At full brightness at neutral white the 24 V side should draw about 3.2 A (3.5 A with 30 stripes).
8. Pair with Zigbee2MQTT (permit join, then follow the device's pairing method on its z2m page). Check that `brightness` and `color_temp` both work from HA. Set `do_not_disturb` **on**: when power returns after an outage the lamp goes back to the state it was in, so a lamp that was off stays off instead of lighting up in the night (with it off, the lamp always turns on).
9. Close the diffuser, run at full power for an hour, and check the strip is comfortably warm, not hot (under ~60 °C), the controller no more than warm (keep it under about 40 °C), and the PSU warm, not hot. Measure the stripe on the PSU on its own.
10. Repeat the lux-meter readings and compare.
11. Fit the wall switch and its automation (`docs/wall-switch.md`).

Tools

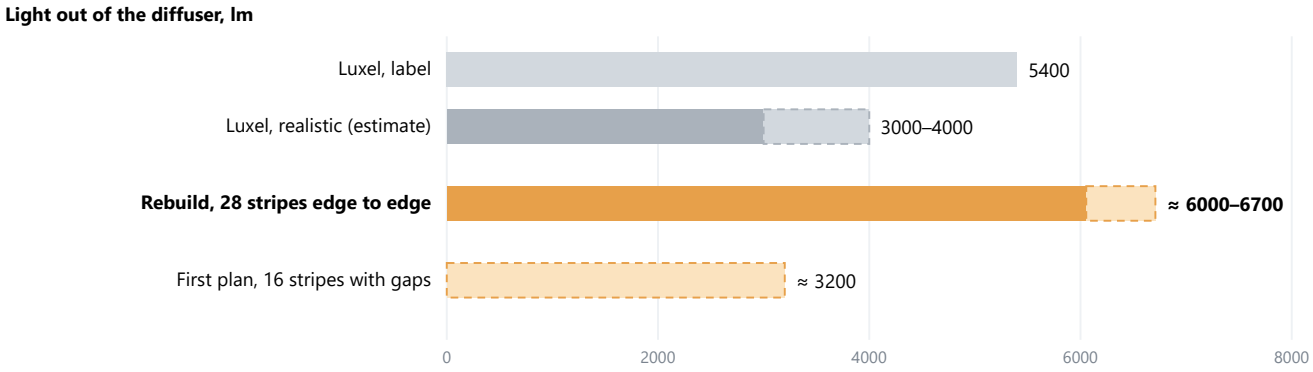
Script	Does
`tools/build_pdf.py`	builds `LedLamp.pdf` from this README and `docs/*.md` with headless Chrome (needs `pip install -r tools/requirements.txt`)

Docs

Doc	About
`docs/brightness.md`	how bright the rebuild will be, against the old lamp, and the numbers behind it
`docs/parts-candidates.md`	every strip, controller and power supply found, with prices and links
`docs/option-b-mains-driver.md`	the fewer-parts option (a mains Zigbee driver) and why it was not chosen
`docs/wall-switch.md`	the wireless rocker switch fixed into the old wall box, and its automation

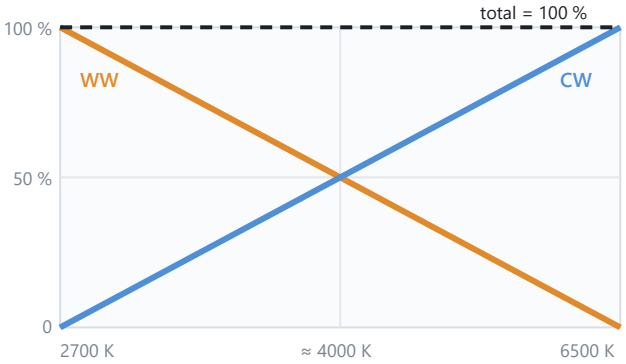
How bright will it be?

About 6000 lm out of the diffuser at any colour temperature, from about 90 W at the wall (the dense layout, 28 stripes edge to edge, ≈ 10.8 m; up to ≈ 6700 lm with 30 stripes).



Any colour temperature. Dashed: estimates, the 30-stripe maximum, or the superseded first plan.

How the numbers are worked out:



channel duty at full brightness

- **Strip:** 1400 lm/m at 14 W/m, but that is **both** channels at 100 %. A Zigbee colour-temperature controller mixes the two whites so they add up to 100 %: at 2700 K only WW is on, at 6500 K only CW, in between both share. So the usable maximum is **one channel's worth: 700 lm/m at 7 W/m**. No Zigbee CCT controller found can change this: MiBoxer confirms its controllers keep 50 % + 50 % at neutral and "it can not be changed".
- **Diffuser:** an opal cover passes about 80 %.
- **Power supply + controller:** about 85 % × 98 % efficient.

	LED power	From the wall	Light out of the diffuser
Rebuild, 28 stripes ≈ 10.8 m, any colour temperature	**76 W**	**≈ 90 W**	**≈ 6000 lm**
Rebuild, 30 stripes ≈ 11.9 m	83 W	≈ 100 W	≈ 6700 lm
First plan, 16 stripes ≈ 5.7 m	40 W	48 W	≈ 3200 lm
Luxel, label	—	72 W	5400 lm
Luxel, realistic guess	—	~45–55 W	~3000–4000 lm

The Luxel's real output: a "72 W" Chinese fitting usually draws 45–55 W, and at 70–75 lm per wall-watt that is **3000–4000 lm** with both colour channels on (neutral). That is a rough estimate, not a measurement. Against it the dense rebuild is **about 1.5–2× as bright at neutral**, and more at the warm and cool ends, where the Luxel likely uses only half its LEDs. For an 18–20 m² living room, ~3000 lm is comfortable general light, so the rebuild will mostly run dimmed, with headroom.

The one Zigbee route to more light at neutral (written for the first plan; with the dense layout it would double the power to ~165 W and isn't needed): two single-channel Zigbee dimmers (e.g. 2× MiBoxer E2-ZR in single-colour mode, one per white) mixed by an HA template light, so both whites can run at 100 %. That gives up to ~6300 lm at neutral from ~80 W, but needs a second E2-ZR (+1210), a 100 W supply, and at 14 W/m the strip wants aluminium under it. Not part of this build.

Measure instead of guessing: put a phone lux-meter app on the floor under the lamp, and write down the old lamp's readings at warm, neutral and cool. Repeat after the rebuild at the same spot.

All candidates found

CCT COB strip, 24 V, 3-wire

Product	Shop	UAH/m	Specs	Notes
LEDTech 24v COB/FCOB CCT 2700–6500K	prom.ua	225	14 W/m, 608 LED/m, Ra>90, 10 mm	**Pick**
LT COB-24-MW-608, 7+7 W/m	svetum.com.ua	216	14 W/m, CRI 90, 10 mm, IP20	Sold in 5 m steps only
RISHANG RM0821MC-L 2700/6500K	prom.ua	213	10 W/m, Ra>90, 8 mm	A third less light per metre
RISHANG RM0831MC-A-1 2700/5700K	prom.ua	181	10 W/m, 950 lm/m	Only to 5700 K
ISOBO V-PRO COB 608 CCT	prom.ua (VOLT)	275	7 W/m, CRI≥90	Half the light per metre
COLORS D8140SWW, 2835 SMD, 19.2 W/m	lightland.com.ua	311	1710 lm/m, CRI 90	Most light per metre, but 1553 per reel breaks the budget; SMD, not COB

Zigbee CCT controllers

Product	Shop	UAH	Specs	Zigbee2MQTT
Moes ZLD-RCW RGB+CCT (pick)	prom.ua (SELLBOT) · Ledenerho, 985	849	12–24 V, 6 A/ch, 90×40×24 mm	ZLD-RCW_1
MiBoxer E2-ZR W+CCT 2-in-1 (first pick)	prom.ua (OPTSVET) · VITRINA, 1342	1210	12–24 V, 12 A, 17.6 mm thick, + RF	E2-ZR
MiBoxer FUT037Z+ 3-in-1 RGB/RGBW/RGB+CCT	OLX, Poltava, private seller, ~807 delivered	679	12–24 V, 6 A/ch, 12 A, 74.5×36×17 mm, 16 kHz, + RF	FUT037Z+
Tuya C02Z (BTF-Lighting and others), dual white	prom.ua (sxemki)	600	5–24 V, 7.5 A/ch, 74.5×35.6×16.5 mm	TS0502B. **Out of stock** on 2026-10-03
LEDLIGHT CCT 10A (Tuya)	prom.ua	670	12–24 V, 53×23×9 mm	**Out of stock** on 2026-10-03
Aqara Dimming Module T1 (ZNDDMK11LM)	prom.ua (SELLBOT)	1599	12/24 V, 75×33×19 mm	ZNDDMK11LM: warmest ≈ 2700 K
Gledopto GL-C-203P (= GL-C-006P)	prom.ua	780	Pro+ CCT	GL-C-006P. **Out of stock**
Sunricher SR-ZG9101CS (single channel)	svl.ua	1578	12–36 V, 8 A	SR-ZG9101CS; only for the two-dimmer route

24 V metal stick power supplies, 100–120 W (dense layout, pick)

Searched on prom.ua, Epicentr and led-story.ua on **2026-10-11**. The supply has to be a **metal stick**, because one stripe is stuck on top of it, and prom.ua Smart delivery was preferred. Rozetka refused automated lookups (HTTP 403); its Diasha 24V 100W listing is probably the same stick as below, with compound filling and soft start, but its size and case couldn't be checked.

Product	Shop	UAH	Size, mm	Notes
Diasha PPF-SL-C-24V-120W "pencil"	prom.ua (EcoLed, 98 %), Smart	320	340×17×18	**Pick**: metal, IP20, 63–70 % load. Spec table says 150 W / 6.25 A, title 120 W / 5 A; no protections or warranty listed, so ask the seller
Diasha PPF-SL-C-24V-100W	prom.ua (Art-Studio, 97 %) · prom.ua (EcoLed),	240	340×17×18	Same stick at 100 W, 1-year warranty (Art-Studio); 76–83 % load, too tight with a stripe heating it

Product	Shop	UAH	Size, mm	Notes
"pencil"	Smart			
PROLUM Series-UTF 100W 24V 4.17 A	Epicentr (free to Epicentr lockers) · prom.ua (ІНФОЛАЙТ), 564	512	264×21.5×20	**Swap**: sealed aluminium, IP67, short-circuit/overload/overheat/overvoltage, 24 months. No Smart
LED STORY Profi 24V 4.1A 100W	led-story.ua	390	317×17×17	Metal, overload/short-circuit/overheat, 2 years; not on prom.ua or Epicentr; stock unclear
LED STORY Profi 100W IP67	led-story.ua	509	317×20×20	Metal, sealed; reaches 60–70 °C at full load per the shop
BIOM STICK BPBLS-72-24	prom.ua (LedDiod, 98 %)	402	313×15×18	BIOM's largest 24 V stick: 72 W covers only ~24 stripes. Two of them (0 V joined, +24 V never joined) would do
LUMINARIA S-24V-72W	prom.ua	395	312×18×18	72 W, too small
Voltronic "PSW24-100"	prom.ua (Brooklyn)	333	312×18×18	**Rejected**: plastic, no short-circuit or overload protection

Epicentr's BIOM sticks go up to 36 W at 24 V only.

24 V power supplies, 100–150 W, not stick-shaped

Searched on prom.ua on **2026-10-11**, after the layout changed to 27–30 stripes laid edge to edge (≈ 11–12 m, at most ≈ 83 W / 3.5 A). Any unit up to about 50 mm wide and 24 mm tall fits in a rim segment outside the strip band.

Product	Shop	UAH	Size, mm	Notes
BIOM BPTH-24-150, 24 V 6.25 A	prom.ua (Електра&Еко, 99 %)	557	154×34×13	Not a stick (picked briefly on 2026-10-11): IP44, short-circuit and overload protection, 24-month warranty; ~55 % load. There is no 100 W BIOM slim
Mi-light SSGL-100-24, 24 V 100 W	prom.ua (OPTSVET, 96 %)	771	145×48×24	Best true 100 W: metal, IP20, 36-month warranty. Title says 4.2 A, spec table says 2.5 A, so ask the seller
BIOM LED-24-100, 24 V 4.2 A	prom.ua (OPTSVET)	201	138×50×23	Budget fallback: BIOM's cheap line, not Professional; protections not listed; 83 % load
Mean Well LPV-100-24, 24 V 4.2 A	prom.ua (База Електрики, 99 %)	1035	190×37×52	Best documented (88 %, short/overload/OVP), but 52 mm tall, so it lies on its side; heavy and pricey
Mean Well LRS-100-24	prom.ua (База Електрики)	675	129×97×30	**Rejected**: 97 mm too wide for a rim segment

Slim 24 V power supplies, 60 W (the old 16-stripe layout)

Product	Shop	UAH	Size, mm	Notes
LED STORY Profi 60W slim	led-story.ua	218	297×17×17	Fallback; house brand, in stock
BIOM STICK BPBLS-60-24	prom.ua (MaxiLight) · prom.ua (LedSpectr), 353	327	313×15×18	First pick, too small for the dense layout; known brand. Out of stock at biom.ua (358), svetum and 5watt on 2026-10-03; samsnab (315) only by Viber inquiry
COXO DS-60-24	lightland	585	130×53×27	CE/CQC, 85 %, 3-year warranty; over budget
EUROLAMP PS 60W 24V 2.5A	eurolamp.ua	469	116×53×27	Out of stock; rated only to +40 °C

Product	Shop	UAH	Size, mm	Notes
No-name XJK-60-24	prom.ua	206	123×34×22	Risky: ripple and inflated rating

Wall switch: classic rockers, for an elderly family member

The lamp's controller must stay powered, so the old mechanical switch goes: its two wires are joined permanently and a **battery Zigbee switch that looks like an ordinary two-key rocker switch** takes its place, **fixed into the old 60 mm box**.

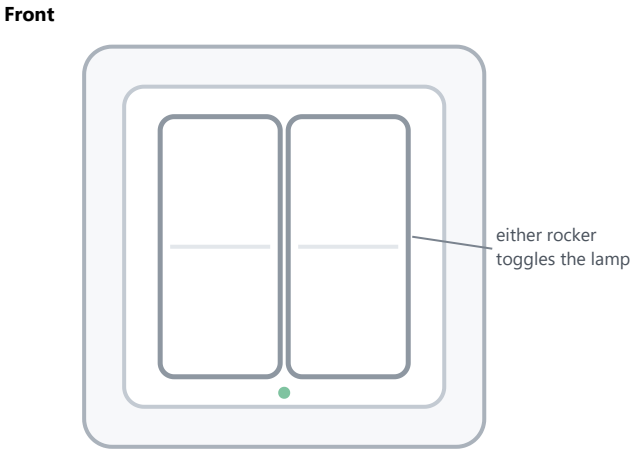
Parts: ≈ 820 UAH

Part	Pick	Shop	Price, UAH
Switch	**Aqara Wireless Remote Switch E1, two rockers** (WXKG17LM): two rockers on an 86×86×15 mm plate, CR2032, Zigbee 3.0	SELLBOT on prom.ua, ready to ship (single-rocker WXKG16LM: 600 at sellbot)	650
Box adapter	**Schneider Asfora blank EPH5600121**, white, 83×83 mm, **fixes into the box with spreading claws**	prom.ua (Електро Крамниця) · schneider.kiev.ua · Epicentr · 5watt, 133	127
Small parts	3M VHB foam tape (~1.1 mm), spare CR2032, 1 **WAGO 221-412** (Електро Крамниця, 23)	local	~40
	Total		**≈ 817**

Order the blank and all the WAGO connectors together. Nearly every prom.ua seller of the blank has a minimum order (150–700 UAH; MegaSnab wants 700). Електро Крамниця asks 200, and the blank (127) plus five genuine WAGO 221-412 (4 needed, 1 spare) comes to 242. Its 15 UAH "Wago 221-412" listing is a PROLUM copy: skip it.

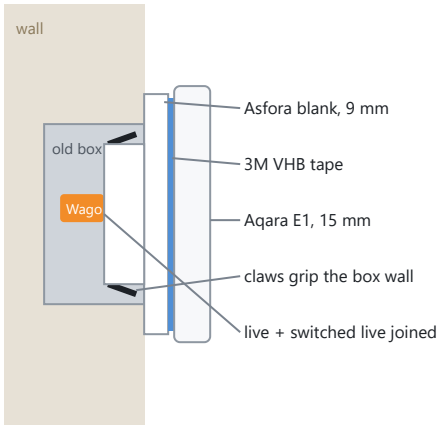
Why these: - **Looks like the switch they know:** rockers you press, like the classic two-key switch, not a flat touch panel or a small button. For only ≈ 50 more than the single-rocker E1 it matches a two-key switch, and **both rockers do the same thing** (toggle the lamp), so whichever side gets pressed, it works. It has a "fast" click mode in which a single press acts at once instead of waiting to see if a double press follows. Z2M: WXKG17LM (single, double and hold per rocker, plus both at once). - **Why the blank plate:** Soviet-era boxes have **no screw lugs**; switches were held by spreading claws. So nothing can screw into the hole. The Asfora blank clamps into the box with its own claws, and the switch is bonded onto its flat plastic face. The blank does the bolting; the switch never relies on crumbly plaster. - **Alternatives:** the Moes 1-button TS0041 panel (690, screw holes, but a flat push button), the single-rocker Aqara E1 WXKG16LM (600), and the Aqara H1 single rocker (binds directly to the lamp, so it works without HA, but it isn't sold in Ukraine; the two-rocker H1 costs ~1299). IKEA RODRET is cheaper but has separate on/off buttons.

Mounting



Aqara E1 two rockers (86 mm) on the Asfora blank (83 mm)

Side section



≈ 25 mm proud of the wall in total

```
 wall          old 60 mm box                    Stack, side view:
                                              wall ▓ │ Asfora blank │ VHB │ Aqara E1
 ▓▓▓
```

Setting it up for an elderly user

- **One action only**: a single press on **either rocker** toggles. Double press and hold do nothing. Don't give the second rocker a different job; two rockers that behave differently are confusing.
- **Always the same light**: "on" is always 100 % at 3000 K, never last night's dim setting.
- **Debounce**: presses within 1 s of the last one are ignored, so a double press by mistake doesn't switch the light straight back off.
- **Mount it exactly where the old switch was**; habit leads them there.
- **Battery**: replace the CR2032 on a schedule (e.g. every 12 months) and add an HA low-battery alert for the caregiver.
- **After a power cut the lamp goes back to how it was** (`do_not_disturb` on, see the build checklist): if it was off it stays off, so it doesn't wake anyone at night. If it was on, it comes back on.
- **The one real risk:** this switch works through HA. If HA or Zigbee2MQTT is down, pressing it does nothing. Keep HA on a UPS, and give the caregiver a second way to switch the lamp (the HA app; the Moes controller has no RF remote).

Option B, fewer parts: a mains Zigbee CCT driver

One box instead of PSU + controller: a **230 V Zigbee constant-current CCT driver** (Tuya `TS0502B` type, e.g. DC 72–185 V, 260 mA, 2 channels with common +) driving **high-voltage 3-pin chandelier strip**, for example the Electro Lab 3 PIN 5B10C/5B20C, 85 UAH/m (3000 + 6500 K, 60 V per metre). It works in Z2M (TS0502B) and could cost about 600–900 UAH, but **no Ukrainian seller of the Zigbee driver was found** (AliExpress/Taobao only), the strip carries up to ~185 V DC on exposed pads, and cheap drivers are often not isolated from mains. Not recommended.
